Software Start-Up Project

Project 2: Hypothesis Validation & MVP Experimentation

CampusPrint - No-Queue Residence/Campus Pickup Printing MVP

Evidence-integrated submission version based on Revised Project 1 v5 Residence Scope Clarification

Team Splendor

Submission note This evidence-integrated version includes actual Google Form response analysis, MVP evidence figures and the final GitHub repository URL for Project 2 technical documentation.	Name	Email
MCS251008	LIU RUNKUN	liurunkun@graduate.utm.my
MCS251009	LUO XI	luoxi@graduate.utm.my
MCS251015	CHU YANQIAO	chuyanqiao@graduate.utm.my
MCS251036	FAHIM KAMAL AHMED	kamalahmed@graduate.utm.my
MCS251006	ZHU TAORUN	zhutaorun@graduate.utm.my

Submission note
This document includes actual Google Form response analysis and evidence figures from the controlled MVP order-flow test. The team should add the final GitHub/GitLab repository URL before submission.

Executive Summary

CampusPrint is a residence/campus-pickup focused printing service concept for UTM KL students who experience long queues, walking distance, USB-related risks and timing pressure when using campus print shops. Project 1 v5 refined the earlier specific-block wording into a safer pilot context: one selected UTM KL residence or nearby student accommodation area, with the exact residence/block name to be confirmed during the pilot test.

Project 2 tests the riskiest assumptions behind this refined model. The MVP is not a full automated platform. It is a concierge-style, low-code MVP where students see a QR poster or WhatsApp message, upload PDF files through Google Form or WhatsApp, review basic black-and-white pricing, show payment proof or payment intent, receive a WhatsApp pickup code, and collect the printed document from a controlled residence/campus pickup point.

The Project 2 learning objective is to determine whether UTM KL students can understand and complete the simulated MVP order-flow, show willingness to submit files online, indicate willingness to reserve or pay, and value no-queue nearby pickup enough to justify a later live Concierge MVP. Evidence was collected through a controlled Google Form MVP order-flow test, payment-intent questions, usability feedback and response-summary analysis.

Important scope correction from Project 1 v5
The Project 2 MVP excludes 24/7 service, door-to-door delivery, color printing, binding, lecturer delivery, club packages, smart lockers, full mobile app and TNG API integration. These remain future scope after the pilot workflow proves demand, willingness to pay and operational feasibility.

Assessment Alignment Checklist

Project 2 required component	Where it is addressed in this report
Part A - Assumption & Hypothesis Extraction	Covered by extracting assumptions from Project 1 v5 BMC, VPC, target scope clarification and MVP risk table.
Part B - Risk Prioritization	Covered by uncertainty-impact-priority mapping focused on simulated order-flow completion, payment intent, online file submission, pickup preference and manual operation.
Part C - Hypothesis Formulation	Covered using precise “We believe that...” hypotheses with metrics and success criteria.
Part D - Experiment Selection & Justification	Covered with selected tools aligned to desirability, feasibility and viability risks; quantity is limited intentionally.
Part E - Experiment Design	Covered with goals, users, procedure, evidence, success/failure indicators and decision logic.
Part F - MVP Development	Covered by low-code MVP architecture, workflow, functional requirements and evidence sections.

Project 2 required component	Where it is addressed in this report
Part G - Code Repository & Development Logs	Covered by repository structure, README requirements, daily log table and deliverable checklist.
Part H - Evidence Collection	Covered by evidence chain, actual Google Form response analysis and interpretation tables.
Part I - Learning Analysis & Interpretation	Covered by validation matrix and decision rules for validated, partially validated or invalidated hypotheses.
Part J - Iteration & Improvement	Covered by keep/revise/postpone actions that feed Project 3 UX refinement.
Part K - Technical & Startup Reflection	Covered by technical and startup reflection linked to Project 1 v5 scoping decisions.

1. Project Overview and Continuity from Project 1 v5

Item	Description
Project name	CampusPrint - No-Queue Residence/Campus Pickup Printing MVP
Broader target market	UTM KL students living in the university residence or nearby student accommodation who need to print assignments, lecture notes, forms and study materials.
MVP pilot / beachhead segment	Students in one selected UTM KL residence or residence/student-accommodation cluster, such as the selected UTM KL residence. This keeps the experiment operationally controllable and produces clearer evidence.
Future expansion segments	Other residences/student-accommodation areas, lecturers, staff, student clubs and societies. These are excluded from the Project 2 MVP validation sample.
Core customer problem	Students lose time and experience stress because campus print shops can be crowded, far from residences/student-accommodation areas, limited by operating hours and dependent on USB or staff-operated workflows.
Core MVP feature	Upload a PDF online, receive manual confirmation, and collect basic black-and-white prints from a nearby controlled pickup point without queueing at a campus print shop.
Main learning objective	Validate whether students will use and pay for a faster no-queue residence/campus pickup printing workflow before building an automated system.

2. MVP Scope

The MVP is designed as a learning instrument. It is deliberately smaller than the long-term product vision so that the team can test demand, payment willingness and operational feasibility within a short course timeline.

MVP scope	Deferred scope	Why
Google Form or WhatsApp PDF submission	Native mobile app or full website	A full platform is not needed to test whether students submit files and orders.
Basic black-and-white academic printing	Color printing and binding	These add pricing and operational complexity and were moved to future scope in Project 1 v5.
Static QR payment, payment screenshot, pre-order or payment-intent tracking	TNG API or automated payment gateway integration	Payment willingness can be tested without waiting for API approval.
Manual confirmation and WhatsApp pickup code	Automated WhatsApp Business API	Manual confirmation is sufficient for the first controlled pilot.
Controlled residence/campus pickup point and service window	24/7 unattended smart-locker pickup	The team must first test peak-period/no-queue demand and operational feasibility.
Post-service feedback survey and evidence logging	Campus-wide launch, lecturer delivery and club bulk packages	The first experiment should test one segment and one core workflow only.

Meaning of "No-Queue"
In this Project 2 report, "no-queue" means users avoid waiting in the campus print-shop queue by submitting files online and collecting prepared documents from a nearby pickup point. It does not mean instant printing, full 24/7 availability or a guaranteed 3-minute pickup.

Part A - Assumption and Hypothesis Extraction

The assumptions below are extracted from the refined Project 1 v5 logic: BMC Version 2, VPC findings, target user scope clarification, revised MVP workflow and future validation metrics. Each assumption is written as a testable risk rather than a fixed truth.

ID	Assumption extracted from Project 1 v5	Linked hypothesis	Risk type
A1	Residence students have a real time and convenience pain when using campus print shops.	H1, H4	Desirability
A2	Students in the pilot residence will submit real PDF files online when the process is simple and privacy is clear.	H2	Desirability / feasibility
A3	Students will place real printing orders, not only say the idea is useful.	H3	Desirability
A4	Students will complete the order-to-pickup workflow through a controlled residence/campus pickup point.	H3, H5	Desirability / feasibility
A5	Students will show willingness to pay through payment proof, QR payment attempt, pre-order or mock-order/payment-intent action.	H6	Viability
A6	The no-queue value proposition is more attractive than the no-USB supporting message for the first segment.	H4	Desirability / channel
A7	The team can manually operate order intake, confirmation, printing, pickup and issue tracking within the defined service window.	H7	Feasibility
A8	Privacy and trust concerns can be reduced through PDF-only upload, a clear privacy statement and file	H2, H5	Desirability / trust

ID	Assumption extracted from Project 1 v5	Linked hypothesis	Risk type
	deletion policy.		

Part B - Risk Prioritization

Risks are prioritized by impact and uncertainty. The first testing cycle focuses on high-impact assumptions that can invalidate the business model if they fail.

Risk / assumption	Uncertainty	Impact	Priority	Reason
Students may not complete the simulated order flow or show serious ordering intent.	High	High	Critical	If students do not complete the order-flow test or show reservation/payment intent, positive opinions are not enough to justify building.
Students may not show payment intent or willingness to pay.	High	High	Critical	Viability depends on receiving money or clear payment-intent behavior.
Students may abandon the online file submission flow.	Medium	High	High	PDF upload and form/WhatsApp submission are core MVP steps.
Students may not complete pickup after ordering.	Medium	High	High	Order completion proves the full no-queue value, not just initial interest.
No-queue may not be the strongest message.	Medium	Medium	Medium	Marketing and positioning should be based on conversion evidence.
The team may not operate the manual workflow reliably.	Medium	High	High	Manual errors or delays reduce trust and block future scaling.
Users may worry about file privacy.	Medium	Medium	Medium	Privacy concerns can reduce file submission, especially for official documents.
Students may want	Medium	Low	Low	These are future

Risk / assumption	Uncertainty	Impact	Priority	Reason
color printing or binding.				features and not required to validate the core B/W academic workflow.
Lecturers, clubs or delivery may be attractive.	Medium	Low for MVP	Low for Project 2	They are future expansion segments and should not distract from the pilot residence test.

Highest-priority test logic
The first validation cycle should not try to prove every feature. It should answer four binary questions: Will students complete the MVP order-flow? Will they show willingness to submit files online? Will they choose a pickup slot? Will they show willingness to reserve or pay?

Part C - Hypothesis Formulation

The following hypotheses use the "We believe that..." structure and define measurable success criteria. They are deliberately specific to the MVP pilot segment rather than all UTM students.

Code	Hypothesis	Success criteria	Evidence
H1 - Problem intensity	We believe that students in the pilot residence experience meaningful queue, distance or waiting-time pain when using campus print shops.	At least 70% of interviewed/observed target users report or demonstrate queue, distance or waiting-time friction during academic printing.	Interview notes, observation records, queue counts, selected quotes
H2 - Online file submission	We believe that students in the pilot residence will submit real PDF printing files through Google Form or WhatsApp when the process is clear and privacy is explained.	At least 10 valid PDF submissions are received within 7 days, and invalid/incomplete submissions remain below 20%.	Google Form records, WhatsApp screenshots, submission error log
H3 - Order-flow completion	We believe that students will complete the simulated CampusPrint order-flow test during the validation period.	At least 10 completed MVP order-flow responses are collected from UTM KL students within the testing period.	Google Form response records, response-summary charts, order-flow completion table
H4 - Value proposition message	We believe that the no-queue residence/campus pickup message will generate stronger conversion than the no-USB message for the first student segment.	The no-queue advertisement has a higher order conversion rate than the no-USB advertisement.	Ad A/B records, link clicks, inquiries, order conversion table
H5 - Workflow completion	We believe that students will complete the upload-to-pickup workflow because it reduces queueing and walking compared with campus print shops.	At least 70% of valid orders are collected at the pickup point, and users report at least 50% waiting/travel-time reduction compared with their usual process.	Pickup checklist, timestamp log, feedback form
H6 - Willingness to pay	We believe that students will show willingness to pay for no-queue residence/campus pickup printing before we build a full automated system.	At least 60% of valid orders include payment proof, QR payment attempt, mock-order click, pre-order, or deposit-style evidence.	Payment screenshots, QR/payment-intent log, pre-order records
H7 - Manual operational feasibility	We believe that the team can structure a clear manual workflow that	Participants can understand the form, pickup slot and	Usability score, issue log, workflow screenshots

Code	Hypothesis	Success criteria	Evidence
	could later support a live Concierge MVP.	confirmation logic; the average ease-of-use score is at least 3.5/5.	
H8 - Satisfaction and repeat potential	We believe that users who complete pickup will be satisfied and show repeat usage potential.	At least 80% of users rate the service 4 or 5 out of 5, and at least 70% say they would use it again during assignment deadlines.	Validation survey, repeat-use question, post-service comments

Part D - Experiment Selection and Justification

The team selects a small number of experiments from the 44 experimentation tools introduced in class. The experiments are chosen because each one matches a specific desirability, feasibility or viability risk. The sequence moves from weaker discovery evidence to stronger behavior evidence.

Experiment	Risk type	Hypotheses	Why this tool is suitable	Evidence produced	Evidence strength
A Day in the Life Observation	Desirability	H1	Observe real printing queues and user behavior instead of relying only on opinions.	Queue length, waiting time, photos if allowed, observation notes	Medium
Customer Follow-up Interview	Desirability / trust	H1, H2	Clarifies the most important pain points and privacy concerns for the pilot segment.	Short notes, quotes, pain frequency table	Weak to medium
Online Ad / Smoke Test	Desirability / channel	H4	Tests whether the value proposition creates real interest before students use the service.	Views, clicks, inquiries, order link visits	Medium
Split Test	Desirability / positioning	H4	Compares “No Queue” versus “No USB” to decide the strongest first message.	Ad A/B conversion rate	Medium
Usability Test of Order Form	Feasibility / UX	H2, H5	Checks whether students can	Task completion, time to complete,	Medium to strong

Experiment	Risk type	Hypotheses	Why this tool is suitable	Evidence produced	Evidence strength
			complete the form without confusion before the live MVP.	confusion points	
Wizard-of-Oz / controlled MVP order-flow test	Desirability / feasibility	H2, H3, H5, H7	Simulates the order workflow through Google Form and manual backend logic without building a full platform or running a live paid printing service.	Completed form responses, file-submission willingness, pickup-slot choices, reservation/payment-intent records, issue logs	Strong
Mock Sale / Payment-Intent Test	Viability	H6	Tests willingness to pay without automated payment integration.	Payment proof, QR attempt, pre-order, mock-order click	Strong
Validation Survey	Product learning	H8	Collects satisfaction, repeat-intention and improvement feedback after real service use.	Ratings, repeat-use answer, open comments	Medium to strong

Part E - Experiment Design

E1. A Day in the Life Observation

Item	Design
Goal	Measure actual printing friction during peak periods before the MVP launch.
Participants / setting	Campus print shop users observed between 5 PM and 7 PM, plus walking-time observation from the pilot residence to the print shop.
Procedure	Record queue length every 15 minutes, estimate waiting time, note visible bottlenecks, record walking time from the pilot residence to the shop, and note whether students use USB or staff-assisted printing.
Evidence to collect	Observation sheet, queue count, time records, walking time record, photos only if allowed.
Success / failure indicator	Success if meaningful waiting, distance or workflow friction is observed. Failure if printing access is consistently fast and nearby, weakening the problem statement.

E2. Customer Follow-up Interview

Item	Design
Goal	Confirm which pains matter most for the pilot residence segment and identify trust concerns before live testing.
Participants	8-12 students from the selected pilot residence or nearby residence/student-accommodation cluster. Lecturers are not part of the core Project 2 sample.
Core questions	How often do you print academic documents? What is frustrating? Would you submit a PDF online? What privacy or payment concern would stop you? Would nearby pickup help you during deadlines?
Evidence to collect	Interview notes, pain ranking, selected quotes, consented screenshots or audio notes.
Success / failure indicator	Success if most participants identify queue, distance, waiting time or file-transfer risk as meaningful. Failure if the problem appears rare or low priority.

E3. Online Ad / Smoke Test and Split Test

Item	Design
Goal	Measure which value proposition attracts more action before or during the MVP.
Ad A	No Queue Printing - Upload your PDF online and collect near your residence or campus pickup point.
Ad B	No USB Needed - Submit your PDF safely through Google Form or WhatsApp.
Channel	Pilot residence WhatsApp group, class group, QR poster near the residence/study area, or Facebook group if available.
Evidence to collect	Views, link clicks, inquiries, form starts, valid submissions, completed orders by ad source.
Success / failure indicator	Success if one message clearly generates higher conversion. If both perform weakly, revise messaging before further scaling.

E4. Usability Test of Order Form

Item	Design
Goal	Check whether a first-time user can complete the order form without help.
Participants	5 pilot users before launch or during early MVP testing.
Task	Open the QR/order link, upload a PDF, select basic B/W settings, choose pickup time, read privacy/payment instructions, and submit.
Evidence to collect	Task completion, time to complete, error points, skipped fields, user comments.
Success / failure indicator	Success if at least 4/5 users complete the form without help and average completion is under 3 minutes. Failure indicates Project 3 UX refinement needs.

E5. Concierge MVP

Item	Design
Goal	Validate real demand and workflow completion using actual printing orders.
Users	10-15 students from the selected pilot residence or

Item	Design
	residence/student-accommodation cluster.
Process	Student reviews MVP concept -> completes Google Form order-flow -> selects B/W print details -> chooses pickup slot -> answers payment/reservation-intent question -> provides usability feedback.
Operating window	Controlled service window, for example 5 PM-9 PM during the test week. The team may adjust but must state the window clearly before launch.
Evidence to collect	Form records, response summary, payment/reservation-intent data, usability score, pickup-slot preference, issue log.
Success / failure indicator	Success if at least 10 valid orders are completed and at least 70% are collected. Failure requires revising the segment, message, pricing or pickup process.

E6. Mock Sale / Payment-Intent Test

Item	Design
Goal	Test business viability without building TNG API or full payment gateway integration.
Payment evidence options	Static TNG QR payment screenshot, QR scan attempt, pre-order, mock-order click, or deposit-style confirmation.
Procedure	Display transparent B/W pricing before confirmation and record whether the user provides proof or takes a payment-intent action.
Evidence to collect	Payment screenshots, mock-order click logs, pre-order sheet, payment-intent column in Google Sheets.
Success / failure indicator	Success if at least 60% of valid orders show payment proof or payment-intent behavior. Failure suggests pricing/value proposition revision.

E7. Validation Survey

Item	Design
Goal	Measure post-use satisfaction, repeat potential and improvement needs.
Participants	Students who completed pickup or attempted to use the service.

Item	Design
Core questions	Satisfaction 1-5, ease of submission 1-5, convenience 1-5, disappointment if service disappeared 1-5, use again yes/no/maybe, acceptable price, improvement suggestion.
Evidence to collect	Survey responses, charts, open comments, repeat-usage intention.
Success / failure indicator	Success if at least 80% rate 4 or 5 and at least 70% indicate repeat-use potential.

Part F - MVP Development

The MVP is a low-code, human-operated service designed to test the value proposition before building automation. The user should experience a clear order flow, while the backend can remain manual for learning.

F1. Core MVP Workflow

1. Student sees QR poster or WhatsApp message in the selected residence/class group.
2. Student opens Google Form or WhatsApp order link.
3. Student uploads PDF only and enters contact, residence/accommodation name, page count, copies and pickup preference.
4. Form displays basic black-and-white pricing and privacy note.
5. Student provides payment proof or takes a tracked payment-intent / mock-order action.
6. Order appears in Google Sheets; team verifies file, payment evidence and pickup slot.
7. Team prints the document using manual B/W printing.
8. Team labels the order with order ID/pickup code.
9. Student receives WhatsApp confirmation and pickup window/code.
10. Student collects the printed document at the controlled pickup point.
11. Student completes validation survey; team records completion, issue and time-saving metrics.

F2. MVP Components

MVP component	Implementation	Evidence to show
Promotion and acquisition	QR poster, WhatsApp message, Ad A/B copy	Ad screenshots, QR/link click count, group post screenshots
Order intake	Google Form or WhatsApp with PDF upload only	Form screenshot, test submission, actual submission records
Order tracking	Google Sheets order log	Order rows, status updates, timestamps
Payment evidence	Static QR payment, payment screenshot, mock-order click or pre-order tracker	Payment proof/intent column and screenshots
Manual printing	Team prints basic B/W PDF documents	Processing checklist, issue log
Pickup	Controlled residence/campus pickup point, pickup code/window	Pickup completion checklist
Feedback	Google Form validation survey	Survey response charts and comments
Evidence storage	GitHub/GitLab docs folder plus evidence appendix	Repository URL, screenshot folder structure, README file

F3. Google Form Fields

Field	Status	Purpose
Full name / nickname	Required	Order identification
WhatsApp number	Required	Manual confirmation and pickup code
Residence/accommodation name or pickup location	Required	Confirm pilot segment and pickup feasibility
PDF file upload	Required	MVP accepts PDF only
Number of copies and pages	Required	Basic B/W price estimate
Basic B/W print setting	Required	Color printing and binding are excluded from MVP
Pickup time preference	Required	Controlled pickup window
Payment proof or payment-intent option	Required for experiment tracking	Tests viability without payment gateway API
Privacy consent	Required	User agrees that files are used only for printing and deleted after completion
How did you hear about us?	Optional	Links orders to Ad A/B or QR poster source

F4. Functional and Non-Functional Requirements

ID	Requirement	Acceptance condition
FR1	The order form must collect WhatsApp contact, PDF file, basic B/W settings, pickup preference and privacy consent.	Each valid submission creates one complete order row.
FR2	The MVP must support PDF upload only.	Non-PDF files are rejected or users are asked to convert to PDF.
FR3	The payment step must record payment proof or payment intent.	Each valid order has a payment proof, QR attempt, mock-order click or pre-order status.
FR4	The team must send WhatsApp confirmation and pickup code.	Each accepted order receives order ID and pickup instruction.
FR5	The team must track order status.	Every order has a status: received, verified, printing, ready, collected, cancelled, or issue.

ID	Requirement	Acceptance condition
NFR1	The order flow should be usable by first-time users.	At least 4/5 usability test users complete the form without help.
NFR2	The MVP must protect user files.	Files are accessed only for printing and deleted after completion.
NFR3	The operation must be auditable.	Order time, confirmation time, pickup time, issue and feedback are recorded.

Part G - Code Repository and Development Logs

Although the MVP is low-code, a GitHub or GitLab repository is still required to show technical organization, experiment transparency and development progress. The repository should not be empty; it should include the prototype/landing page, README, screenshots and testing documents.

Repository URL: https://github.com/luox62620-crypto/campusprint-mvp-validation

The repository is used as the technical evidence space for README documentation, experiment plan, Google Form screenshots, anonymized evidence analysis, Project 2 report files and optional prototype/landing-page assets.

Repository item	Required content
Repository URL	https://github.com/luox62620-crypto/campusprint-mvp-validation
README.md	Project description, team members, MVP scope, testing workflow, how to access the form/prototype.
/prototype or index.html	Simple landing page or prototype page with value proposition and order link.
/forms	Google Form screenshots, form field design, privacy statement.
/ads	Ad A/B images and message copy.
/docs	Experiment plan, interview questions, observation sheet, validation survey, risk plan.
/logs	Daily development and testing logs.
/evidence	Final screenshots and anonymized order/survey evidence for submission.
/analysis	Spreadsheet exports, charts, hypothesis interpretation and decision tables.

Development Log Template

Day	Development / testing activity	Output / evidence	Responsible role
Day 1	Finalize Project 2 hypotheses, MVP scope and operating window.	Hypothesis table, workflow diagram, service-window decision.	All members
Day 2	Create Google Form, privacy note and Google Sheets order log.	Form screenshot, test submission, order sheet.	Technical + operations members
Day 3	Prepare Ad A/B and QR	No-queue and no-USB ad	Marketing/research

Day	Development / testing activity	Output / evidence	Responsible role
	poster.	screenshots.	members
Day 4	Run usability test and internal order simulation.	Usability notes, issue list, corrected form fields.	All members
Day 5	Launch Concierge MVP in the pilot residence.	Order records, payment-intent records, pickup plan.	Operations members
Day 6	Continue MVP, run split test and collect survey responses.	Ad conversion table, survey charts, pickup records.	Research + operations members
Day 7	Analyze evidence and write final learning/iteration decisions.	Hypothesis validation table, iteration table, Project 3 UX issues.	All members

Part H - Evidence Collection

The team conducted a controlled MVP order-flow test using Google Form. A total of 12 completed responses were collected. The test did not operate a full live printing service. Instead, it simulated the core CampusPrint workflow: consent, participant classification, online file-submission willingness, black-and-white print details, pickup slot selection, payment/reservation intent, and usability feedback.

This evidence should therefore be interpreted as simulated workflow-completion and payment-intent evidence, not as live printing or actual paid-order evidence. It is still useful for Project 2 because it directly tests whether users can understand the order flow, show interest in online submission, choose a pickup slot, and indicate willingness to reserve or pay before a full system is built.

Table H1. Evidence Result and Validation Summary

Evidence item	Target	Actual result	Validation status	Interpretation
Completed MVP order-flow responses	5-8 responses	12 completed responses	Validated	The minimum response target was exceeded, showing enough early evidence for a controlled MVP test.
Residence / nearby accommodation sample	Majority of responses from residence / nearby users	8/12 = 66.7%	Validated	Most responses came from the intended pilot context. Non-residence responses are treated as

				supporting evidence.
Online file submission willingness	>=70% Yes or Maybe	12/12 Yes or Maybe; 9/12 Yes = 75.0%	Validated	Users did not reject online file submission, but privacy concerns remain important.
Payment / reservation intent	>=60% Yes or Maybe	12/12 Yes or Maybe; 9/12 Yes = 75.0%	Validated	The test provides payment-intent evidence without requiring real payment gateway integration.
Mock reservation action	>=60% would reserve	11/12 = 91.7% would click Reserve Slot	Validated	Most participants were willing to take a reservation-style action.
Ease of order-flow test	Average >=4/5 or >=70% rating 4-5	Average 3.75/5; 7/12 = 58.3% rating 4-5	Partially validated	The flow is understandable, but not yet smooth enough. The prototype should be simplified.
Deadline usage potential	>=70% Yes or Maybe	12/12 Yes or Maybe; 8/12 Yes = 66.7%	Validated for deadline use	Users see value during assignment deadlines, but daily usage still needs further testing.
Product attachment / disappointment	>=80% rating 4 or 5	7/12 = 58.3% rating 4 or 5	Not fully validated	The idea has interest, but emotional attachment is not strong enough yet.
Key user concerns	Identify top blockers	File privacy 7/12; pickup reliability 7/12; price concern 5/12	Actionable findings	These concerns directly inform Project 3 UX refinement and next MVP iteration.
Pickup slot preference	Identify realistic operating window	5:00-6:00 PM selected by 5/12	Validated direction	A fixed pickup window is more realistic than 24/7 or instant pickup at this stage.

Table H2. Evidence Categories and Appendix Locations

Evidence category	What was collected	Appendix location	Status
Google Form MVP tool	Form overview, consent, order-flow, payment-intent	Appendix H	Completed - documented with form evidence figures

	sections		
Response summary	12 completed Google Form responses and summary charts	Appendix H	Completed
MVP usage evidence	Form completion, online submission willingness, pickup-slot selection	Part H and Appendix H	Completed
Payment-intent evidence	Willingness to pay/reserve and mock reserve-slot action	Part H and Appendix H	Completed
Usability evidence	Ease score and open-ended improvement comments	Part I and Part J	Completed
Issue evidence	Privacy, pickup reliability, price and upload concerns	Part J	Completed
Real printing order evidence	Actual paid orders and physical pickup records	Future test cycle	Not collected in this controlled simulation

Part I - Learning Analysis and Interpretation

The findings were interpreted using the Project 2 hypothesis-validation logic. Each hypothesis is classified as validated, partially validated, or not fully validated based on the actual Google Form evidence. The team did not treat positive opinions alone as strong evidence; the analysis focuses on completed form actions, payment/reservation intent, and usability feedback.

Table I1. Hypothesis Validation Analysis

Hypothesis	Evidence used	Result	Interpretation	Decision
H1: We believe that UTM KL students have printing pain related to distance, queues, or convenience.	Pain-point question	Walking/distance 6/12; long queue 3/12	Validated	Keep no-queue nearby residence/campus pickup as the main value proposition.
H2: We believe that students are willing to submit files online.	Online submission question	9/12 Yes; 3/12 Maybe; 0 No	Validated	Keep online file submission as the core MVP workflow, but improve trust wording.
H3: We believe that users will complete the MVP order-flow simulation.	Completed Google Form responses	12 completed responses	Validated for simulated MVP	Use this as workflow-completion evidence, not as real paid order evidence.
H4: We believe that the order flow is	Ease score	Average 3.75/5; 7/12 rated 4 or 5	Partially validated	Simplify the next prototype into a

easy enough to use.				clearer 3-step flow.
H5: We believe that students show willingness to pay or reserve.	Payment intent and mock reserve question	9/12 Yes to pay/reserve; 11/12 would click Reserve Slot	Validated	Keep payment intent as viability evidence; avoid full payment gateway for now.
H6: We believe that deadline-period usage potential exists.	Deadline usage and disappointment score	12/12 Yes or Maybe for deadlines; 7/12 high disappointment score	Partially validated	Position CampusPrint around deadline and urgent academic printing first.
H7: We believe that fixed pickup windows are more realistic than 24/7 service.	Pickup-slot choice	5:00-6:00 PM was the most selected slot	Validated	Use fixed pickup windows instead of 24/7 or 3-minute pickup promises.

Overall learning: the MVP test supports the core direction of online submission plus nearby residence/campus pickup. However, the team should not claim that live printing demand or real payment behavior has been fully validated yet. The current evidence validates interest, workflow completion, reservation intent, and UX issues. A later Concierge MVP can test actual paid orders and physical pickup completion.

Part J - Iteration and Improvement

The MVP should now be refined based on the response data. The changes below are evidence-based and should be carried into Project 3 UX refinement, especially the before-after wireframes and clickable prototype.

Table J1. Evidence-Based MVP and Prototype Iteration Decisions

Finding / issue	Evidence source	Decision	Prototype / MVP change	Status
File privacy is a major blocker.	7/12 selected file privacy concern.	Revise	Add a clear privacy statement before upload: files are used only for printing and deleted after completion.	Use in Project 3
Pickup reliability is a major concern.	7/12 selected pickup reliability concern.	Revise	Add order status steps: Submitted, Confirmed, Ready for Pickup, Collected.	Use in Project 3
Price concern affects adoption.	5/12 selected price concern; comments requested reasonable pricing.	Revise	Show estimated total price before the reserve action.	Use in Project 3
The order flow is not	Average ease score	Revise	Simplify the user	Use in Project 3

yet smooth enough.	3.75/5.		journey into three steps: Upload File -> Print Details & Price -> Pickup/Reserve Confirmation.	
Fixed pickup slots are more realistic than 24/7.	5:00-6:00 PM was the most selected slot.	Keep / refine	Use selected pickup windows instead of 24/7 or instant pickup promises.	Keep
Black-and-white printing should remain the MVP scope.	MVP scope and feedback; color requests are future needs only.	Keep / postpone	Keep B/W only in the MVP and move color printing/binding to future scope.	Keep
The MVP is a controlled simulation, not a full service.	No actual paid orders or physical pickup were executed in this validation cycle.	Clarify	State clearly that the current test validates order-flow behavior and payment intent, while actual paid orders require a later Concierge MVP.	Completed

Decision summary: CampusPrint should continue with a focused residence/campus pickup MVP, but the next prototype must improve privacy assurance, pickup status visibility, pricing transparency, and step-by-step clarity. These changes provide a direct input for Project 3 UX refinement.

Part K - Technical and Startup Reflection

Technical Reflection

The CampusPrint Project 2 MVP intentionally uses Google Form, Google Sheets, WhatsApp, static QR/payment-intent evidence and manual workflow tracking instead of a full software platform. This technical decision is appropriate because the key uncertainty is not whether a printing app can be built. The current validation cycle focuses on whether students understand the workflow, show willingness to submit files, indicate willingness to reserve/pay, and find the nearby pickup concept useful.

The technical learning focus is therefore workflow reliability. Even with a manual backend, the user-facing process must be structured: PDF-only upload, clear B/W pricing, privacy consent, payment proof or payment-intent capture, pickup code and order status. The most important technical artifact is not only code, but the evidence infrastructure: order logs, timestamps, issue logs, screenshots, survey charts and repository documentation.

Future technical development should only follow validated learning. If a later live Concierge MVP achieves actual paid orders and pickup completion, the next technical layer can include a secure web ordering page, temporary file storage, automated status notification, automated payment integration, admin dashboard and eventually controlled pickup hardware. If demand is weak, the team should revise the segment or value proposition before building more software.

Startup Reflection

Project 1 v5 corrected the original risk of overbuilding. The original idea mixed many features: 24/7 printing, smart lockers, color printing, binding, delivery, lecturers, clubs, TNG API and campus-wide deployment. Project 2 now follows a more disciplined startup logic: test one core workflow with one pilot segment before expanding.

The most important startup learning is that positive opinions are not enough. The MVP must generate stronger evidence: real file submissions, completed orders, payment proof or payment-intent behavior, pickup completion, user satisfaction and repeat-use intention. These behaviors are more useful than asking whether students “like” the idea.

If the MVP succeeds, CampusPrint can move toward Project 3 by refining the user flow, improving the order form, clarifying pickup instructions, strengthening the privacy/payment screens and preparing a clickable prototype. If it fails, the team can pivot quickly by changing the pilot segment, price, service window, message or pickup process before wasting time on a large system.

7-Day MVP Testing Timeline

Day	Main activity	Expected output
Monday	Finalize hypotheses, service window, Google Form, payment-intent method and evidence checklist.	MVP workflow, form draft, payment-intent tracker.
Tuesday	Run pilot-residence follow-up interviews and form usability test.	Interview notes, usability issues, revised form.
Wednesday	Run A Day in the Life observation and launch Ad A/B smoke test.	Observation sheet, QR/link clicks, inquiries.
Thursday	Launch controlled MVP order-flow test and collect first responses.	Google Form response records, payment-intent evidence, usability notes.
Friday	Continue Concierge MVP and compare no-queue vs no-USB conversion.	Ad conversion table and updated order log.
Saturday	Collect validation survey and complete pickup/issue analysis.	Survey charts, issue log, completion rate.
Sunday	Interpret hypotheses, write learning analysis and prepare Project 3 UX refinement inputs.	Validated/invalidated table, iteration plan, evidence appendix.

Risk Mitigation Plan

Risk	Impact	Mitigation action
Low order volume	Weak demand evidence.	Improve QR/poster message, increase pilot residence exposure, extend test period, or test another residence/student-accommodation cluster.
Low payment intent	Business viability remains uncertain.	Test lower price point, clearer benefit, bundle pricing, or stronger no-queue deadline message.
File privacy concern	Students may not upload documents.	Add privacy notice, PDF-only rule, deletion policy, restricted file access and optional test file first.
Pickup confusion	Orders may not be collected.	Send clear pickup code, map/photo instruction, reminder message and fixed pickup slots.
Printing delay or wrong order	Trust decreases and satisfaction drops.	Use accepted-order limit, status tracker, operator checklist and redo/refund policy.
Weak ad performance	Value proposition may be unclear.	Test shorter headline, deadline-focused copy, no-queue vs no-USB message, and poster position.
Unclear 24/7 demand	Team may overpromise.	Only test defined service windows first; future 24/7 requires separate evidence.
Requests for color/binding/delivery	Scope creep.	Record as future demand but do not add during Project 2 MVP.

Conclusion

This Project 2 report translates Revised Project 1 v5 into a focused evidence-driven validation plan. The report avoids the earlier contradiction of trying to test a campus-wide, 24/7, feature-heavy printing platform. Instead, it tests the smallest meaningful workflow: online PDF submission intention, payment/reservation intent, basic B/W printing preference, pickup-slot selection and controlled residence/campus pickup.

The strongest evidence in this validation cycle comes from actual user behavior inside the MVP test: completed form responses, file-submission willingness, reservation/payment intent, pickup-slot selection and repeat-use intention. These results will determine whether CampusPrint should proceed to a later live Concierge MVP, revise the workflow, or pivot. They also provide evidence for Project 3 UX iteration and the final pitch/demo.

Appendices - Evidence Instruments and Supporting Materials

Appendix A - Pilot User Interview Questions

- How often do you print academic documents such as assignments, notes, reports or thesis materials?
- Where do you usually print, and how long does the whole process usually take?
- What is the most frustrating part: queueing, walking distance, shop hours, USB/file transfer, pricing or something else?
- Would you submit a PDF through Google Form or WhatsApp if the privacy process is clear?
- Would you prefer residence/campus pickup during deadline periods?
- What price would feel acceptable for basic B/W residence/campus pickup printing?
- What concern would stop you from using this MVP?
- Would you use this service again if the first order worked well? Why or why not?

Appendix B - Observation Sheet Template

Time slot	Queue length	Estimated waiting time	Observed issue	Notes
5:00-5:15 PM				
5:15-5:30 PM				
5:30-5:45 PM				
5:45-6:00 PM				
6:00-6:15 PM				
6:15-6:30 PM				
6:30-6:45 PM				
6:45-7:00 PM				

Appendix C - Ad Copy Instruments

Ad version	Draft message
Ad A - No Queue	Deadline printing? Upload your PDF online and collect near your residence or campus pickup point. No campus print-shop queue.
Ad B - No USB	No USB needed. Submit your PDF safely through Google Form or WhatsApp and collect your print near your residence or campus pickup point.

Appendix D - Order Log Template

Order ID	Time received	User code	Pages/ copies	Payment proof/intent	Status	Pickup time	Issue
CP-001							
CP-002							
CP-003							

Appendix E - Validation Survey Questions

- How satisfied are you with CampusPrint? (1-5)
- How easy was it to submit your PDF? (1-5)
- How convenient was the pickup point? (1-5)
- How much time did this save compared with your usual printing process?
- How disappointed would you be if this service disappeared tomorrow? (1-5)
- Would you use CampusPrint again during assignment deadlines? (Yes/No/Maybe)
- Would you recommend CampusPrint to a friend? (Yes/No/Maybe)
- What price would be acceptable for this service?
- What was the best part of the service?
- What should be improved in the next version?
- Did you have any privacy, payment or trust concerns?

Appendix F - Privacy Statement Instrument

Privacy statement for MVP order form
Uploaded files will only be used for printing your order during the MVP test. Files will be accessed only by the team member processing the order and will be deleted after order completion or at the end of the test period. Please do not upload highly sensitive documents if you are uncomfortable with manual handling during the MVP stage.

Appendix G - Evidence Insertion Checklist

Evidence item	Completed?	Appendix / folder location
Observation records	Not executed in this minimum validation cycle	Appendix B provides the observation template; current actual evidence comes from Google Form order-flow testing.
Pilot interview notes	Background evidence only	Project 1 interview evidence informed the MVP scope; Project 2 actual evidence is the Google Form MVP test.
Ad A/B screenshots and performance	Instrument prepared; not executed as actual A/B test	Appendix C contains draft ad copy. No ad performance data was claimed in Part H.

Evidence item	Completed?	Appendix / folder location
Usability test notes	Completed	Part H / Part I / Part J; usability evidence from ease score, user concerns and open-ended improvement comments.
Google Form screenshots and submissions	Completed	Appendix H; repository folder evidence/screenshots and evidence/data.
WhatsApp order screenshots	Not used in this controlled test	The MVP test used Google Form rather than WhatsApp order screenshots to avoid claiming live printing orders.
Payment proof or payment-intent records	Completed as payment-intent evidence	Part H; repository folder evidence/data; no real payment collection was claimed.
Pickup checklist and issue log	Simulated only	Pickup slot preference and pickup reliability concern captured in Part H; physical pickup checklist deferred to next Concierge MVP cycle.
Validation survey charts	Completed	Appendix H Figure H5; repository folder evidence/screenshots and evidence/data.
GitHub/GitLab repository link	Completed	https://github.com/luox62620-crypto/campusprint-mvp-validation

This checklist distinguishes completed Google Form MVP evidence from tools/templates that were prepared but not executed during the minimum validation cycle.

Appendix H - Google Form MVP Test Evidence Figures

The following figures document the Google Form MVP order-flow test and summarize the actual response data collected from 12 participants. Figures H1-H4 show the form sections used for the controlled MVP test. Figure H5 summarizes the Google Form response data used in Part H, Part I, and Part J.

Figure H1. Google Form MVP test overview.

CampusPrint MVP Order-Flow Test

This is an MVP validation test for the Software Start-Up Project. No actual printing or payment will happen unless separately confirmed. Participants should not upload sensitive files. The test simulates online file submission, pickup-slot selection, payment/reservation intent, and usability feedback.

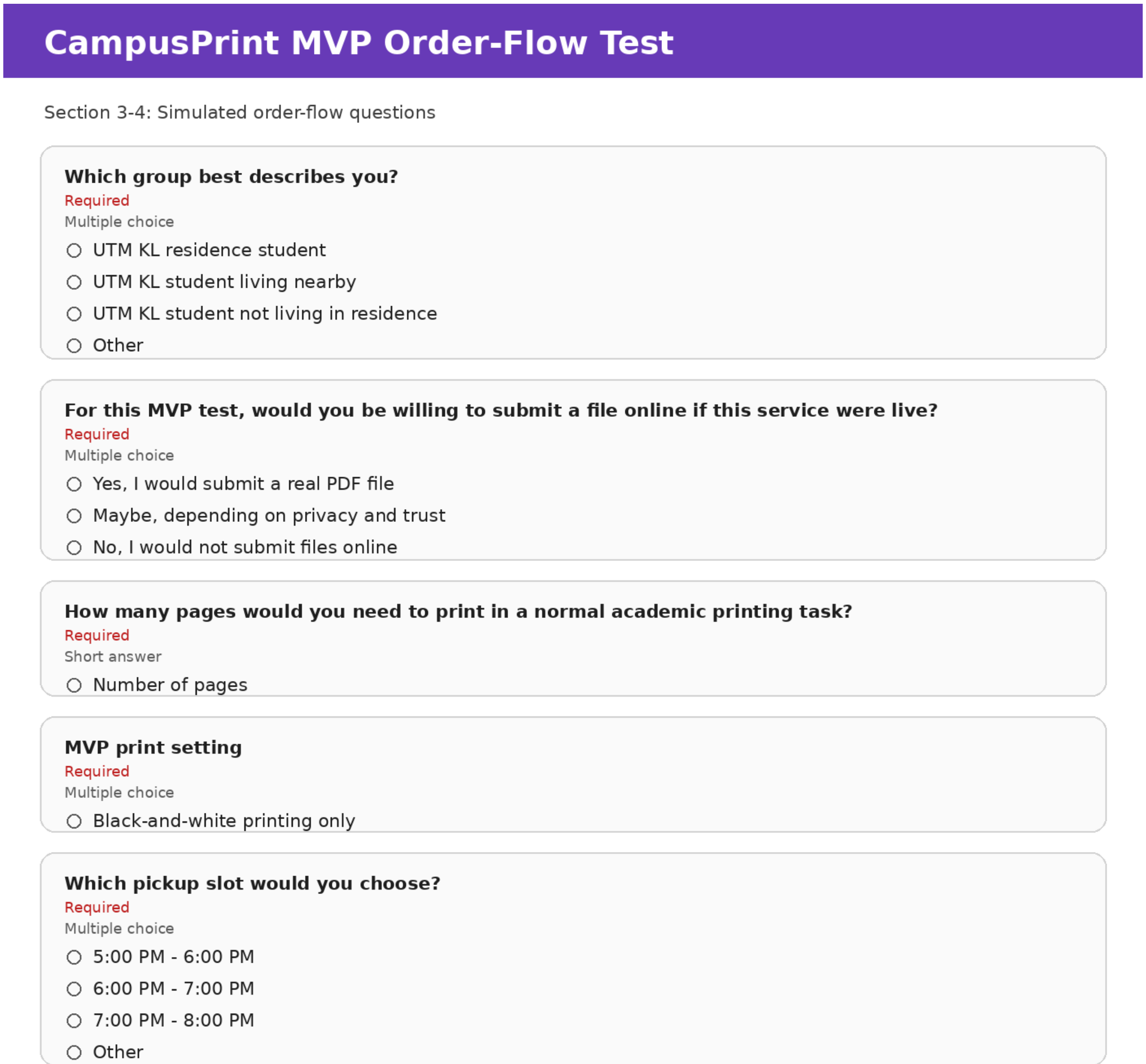
Figure H2. Consent question used before the MVP order-flow test.

CampusPrint MVP Order-Flow Test

Section 1: Consent

I understand this is an MVP validation test. No actual printing or payment will happen unless confirmed by the team.
Required
Checkbox
○ I understand and agree to participate.

Figure H3. Simulated order-flow questions: participant type, online file submission, B/W print details, and pickup slot.



CampusPrint MVP Order-Flow Test

Section 3-4: Simulated order-flow questions

Which group best describes you?
Required
Multiple choice
- ○ UTM KL residence student
- ○ UTM KL student living nearby
- ○ UTM KL student not living in residence
- ○ Other

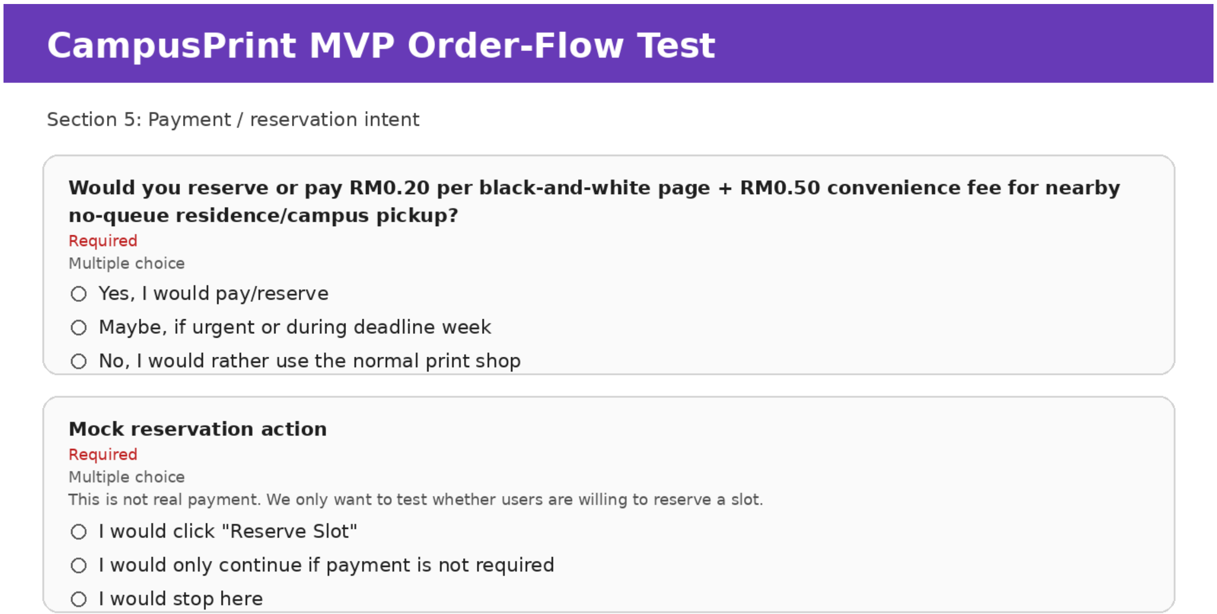
For this MVP test, would you be willing to submit a file online if this service were live?
Required
Multiple choice
- ○ Yes, I would submit a real PDF file
- ○ Maybe, depending on privacy and trust
- ○ No, I would not submit files online

How many pages would you need to print in a normal academic printing task?
Required
Short answer
- ○ Number of pages

MVP print setting
Required
Multiple choice
- ○ Black-and-white printing only

Which pickup slot would you choose?
Required
Multiple choice
- ○ 5:00 PM - 6:00 PM
- ○ 6:00 PM - 7:00 PM
- ○ 7:00 PM - 8:00 PM
- ○ Other

Figure H4. Payment and reservation-intent questions.



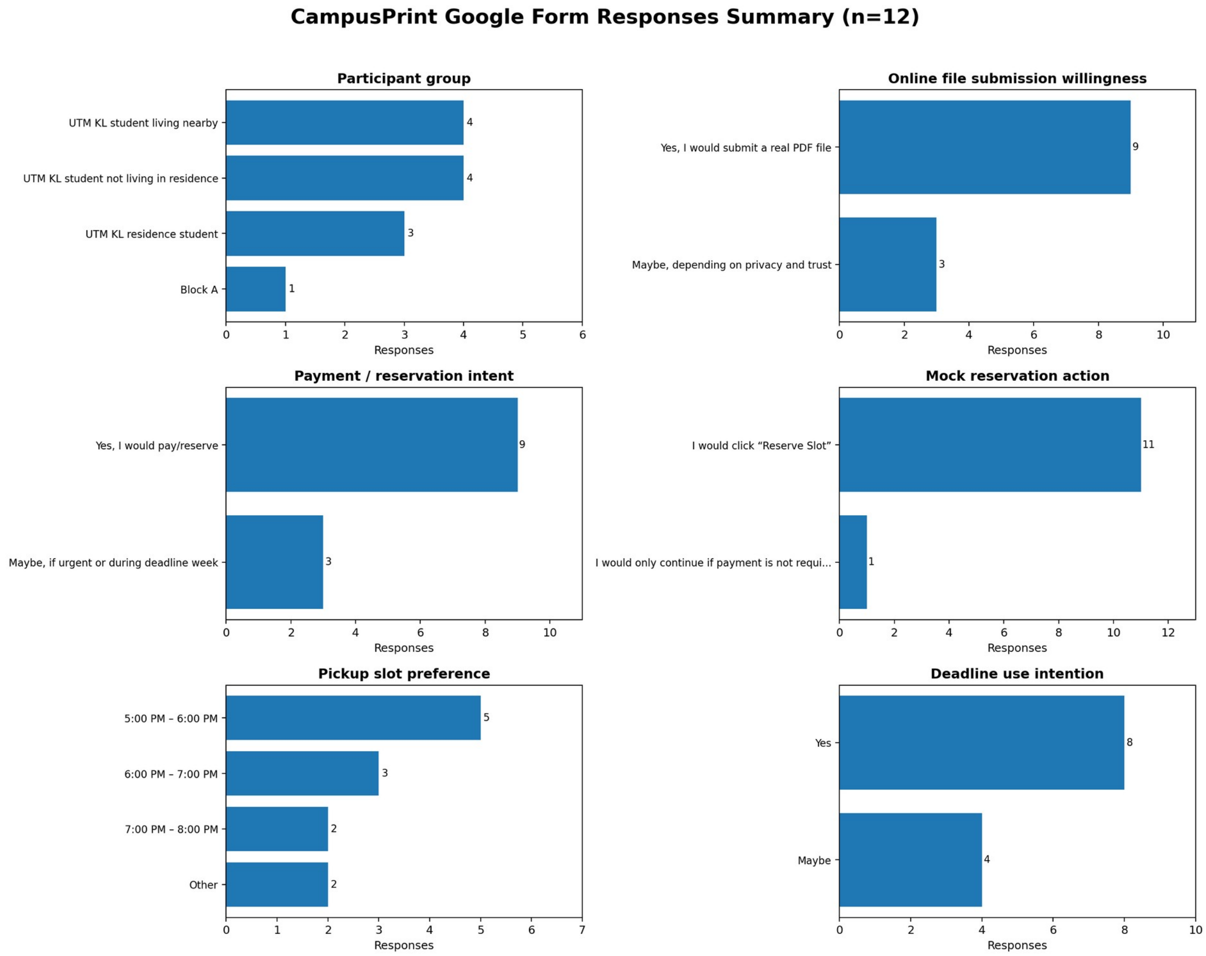
CampusPrint MVP Order-Flow Test

Section 5: Payment / reservation intent

Would you reserve or pay RM0.20 per black-and-white page + RM0.50 convenience fee for nearby no-queue residence/campus pickup?
Required
Multiple choice
- ○ Yes, I would pay/reserve
- ○ Maybe, if urgent or during deadline week
- ○ No, I would rather use the normal print shop

Mock reservation action
Required
Multiple choice
This is not real payment. We only want to test whether users are willing to reserve a slot.
- ○ I would click "Reserve Slot"
- ○ I would only continue if payment is not required
- ○ I would stop here

Figure H5. Google Form response summary generated from the CSV responses.


CampusPrint Google Form Responses Summary (n=12)
Participant group
UTM KL student living nearby
UTM KL student not living in residence
UTM KL residence student
Block A
4
4
3
1
Responses
Online file submission willingness
Yes, I would submit a real PDF file
Maybe, depending on privacy and trust
9
3
Responses
Payment / reservation intent
Yes, I would pay/reserve
Maybe, if urgent or during deadline week
9
3
Responses
Mock reservation action
I would click "Reserve Slot"
I would only continue if payment is not requi...
11
1
Responses
Pickup slot preference
5:00 PM – 6:00 PM
6:00 PM – 7:00 PM
7:00 PM – 8:00 PM
Other
5
3
2
2
Responses
Deadline use intention
Yes
Maybe
8
4
Responses